**JECRC UNIVERSITY, JAIPUR**
**Session: Jan-June 2026**
**Subject: Mathematics-II (DMA025A)**
**B. Tech. II Semester I Year 2025-26**
**Tutorial Sheet–1 (M.M: 64)**

## UNIT-1: Matrices

**CO3:**

### PART – A (5*2=10)

**A1:** Define Power series.

**A2:** Define ordinary point.

**A3:** Define singular point.

**A4:** Define Bessel's Equation.

**A5:** Define Legendre's Equation.

### PART – B (3*7=21)

**B1:** Find the power series solution $(1-x^2)y'' - 2xy' + 2y = 0$.

**Ans**. $y = a_0\left[1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} + \cdots\right] + a_1 x.$

**B2:** Solve $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx} - y = 0$.

**Ans.** $y = a_0\left[1 + \frac{x^2}{2} - \frac{x^4}{8} \ldots\right] + a_1 x.$

**B3: Solve** $y'' + xy' - x^2y = 0$. Ans. $y = a_0\left[1 - \frac{x^4}{12} - \cdots\right] + a_1\left[x - \frac{x^3}{6} - \frac{1}{40}x^5 \ldots\right]$

### PART – C (3*11=33)

**C1:** Solve in series the equation $\frac{d^2y}{dx^2} + x^2y = 0$.

**Ans.:** $y = a_0\left(1 - \frac{1}{12}x^4 + \frac{x^8}{12\times7\times8} - \frac{x^{12}}{12\times7\times8\times11\times12} + \cdots\right) + a_1\left(x - \frac{x^5}{20} + \frac{x^9}{20\times8\times9} \ldots\right)$

**C2:** Solve the Bessel's equation $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (x^2 - n^2)y = 0$.

**C3:** Solve the Legendre's equation $(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + n(n+1)y = 0$.

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